CSA5108 Cryptography and Network security

Exp 22 : CBC mode using S-DES
Code:

```
#include <stdio.h>

int E(int x, int k) {
   return x ^ k;
}

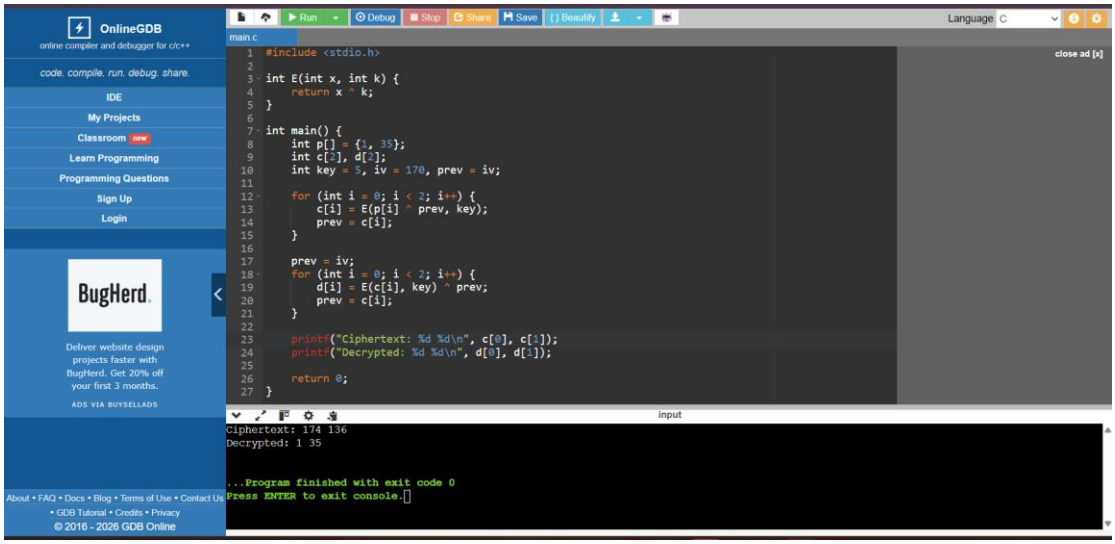
int main() {
   int p[] = {1, 35};
   int c[2], d[2];
   int key = 5, iv = 170, prev = iv;

   for (int i = 0; i < 2; i++) {
      c[i] = E(p[i] ^ prev, key);
      prev = c[i];
   }

   prev = iv;
   for (int i = 0; i < 2; i++) {
      d[i] = E(c[i], key) ^ prev;
      prev = c[i];
   }

   printf("Ciphertext: %d %d\n", c[0], c[1]);
   printf("Decrypted: %d %d\n", d[0], d[1]);

   return 0;
}
```


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#include <stdio.h>
int E(int x, int k) {
return x ^ k;
}
int main() {
int p[] = {1, 35};
int c[2], d[2];
int key = 5, iv = 170, prev = iv;
for (int i = 0; i < 2; i++) {
c[i] = E(p[i] ^ prev, key);
prev = c[i];
}
prev = iv;
for (int i = 0; i < 2; i++) {
d[i] = E(c[i], key) ^ prev;
prev = c[i];
}
printf("Ciphertext: %d %d\n", c[0], c[1]);
printf("Decrypted: %d %d\n", d[0], d[1]);
return 0;
}
input
Ciphertext: 174 136
Decrypted: 1 35
...Program finished with exit code 0
Press ENTER to exit console.